

HTML Notes

Introduction & Basic Structure

1. What is HTML?

- HTML stands for **HyperText Markup Language**.
- It is used to create and structure web pages.
- HTML defines the content and structure of a webpage, such as headings, paragraphs, links, images, tables and forms.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Hello World</h1>
  <p>Welcome to HTML.</p>
</body>
</html>
```

★ Benefits of HTML:

- ✓ Simple and beginner-friendly.
- ✓ Provides the structure of web pages.
- ✓ Works with CSS for styling and JavaScript for behavior.
- ✓ Supported by all modern web browsers.

2. Basic HTML Document Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  Page content goes here
</body>
</html>
```

Part	Purpose
<!DOCTYPE html>	Declares the HTML document type
<html>	Root element of the document
<head>	Contains metadata and page information
<title>	Sets the browser tab title
<body>	Contains visible webpage content

3. HTML Tags, Elements & Attributes

HTML uses **tags** to mark and structure content. An HTML element normally consists of an opening tag, content and a closing tag.

```
<p>Hello World</p>
```

Opening tag → <p>

Content → Hello World

Closing tag → </p>

HTML Attributes: Attributes provide additional information about an element and are written inside the opening tag.

```
<a href="https://example.com">Visit</a>
```

```

```

```
<p id="intro" class="text">Welcome</p>
```

Attribute	Used With	Purpose	Example
href	<a>	Specifies link destination	href="page.html"
src		Specifies image path	src="pic.jpg"
alt		Alternative text for image	alt="Profile"
id	Most elements	Unique identifier	id="main"
class	Most elements	Groups elements for CSS/JS	class="box"
title	Most elements	Extra information/tooltip	title="Info"

4. Headings & Paragraphs

```
<h1>Main Heading</h1>
```

```
<h2>Sub Heading</h2>
```

```
<h3>Section</h3>
```

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

HTML provides headings from **<h1>** to **<h6>**. Paragraphs are written using **<p>**.

★ **Note:** Use headings according to the document structure. Do not use headings only to make text larger; CSS controls visual size.

5. Text Formatting

```
<b>Bold</b>
```

```
<strong>Important text</strong>
```

```
<i>Italic</i>
```

```
<em>Emphasized text</em>
```

```
<u>Underline</u>
```

```
<mark>Highlighted</mark>
```

```
<small>Small text</small>
```

```
<del>Deleted text</del>
```

```
<ins>Inserted text</ins>
```

```
<sup>2</sup>
```

```
<sub>2</sub>
```

6. Links & Images

Links: The `<a>` tag creates hyperlinks. The **href** attribute specifies the destination.

```
<a href="https://example.com">Visit Website</a>
<a href="about.html">About Us</a>
<a href="#contact">Go to Contact</a>
```

Images: The `<img>` element displays an image. It commonly uses **src** and **alt** attributes.

```

```

★ **Note:** The **alt** attribute provides alternative text when the image cannot be displayed and helps accessibility.

7. HTML Lists

Unordered List:

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
  <li>Mango</li>
</ul>
```

Ordered List:

```
<ol>
  <li>Login</li>
  <li>Select Product</li>
  <li>Checkout</li>
</ol>
```

Description List:

```
<dl>
  <dt>HTML</dt>
  <dd>Structures web pages.</dd>
</dl>
```

Tag	Purpose
	Unordered list
	Ordered list
	List item
<dl>	Description list
<dt>	Term/name
<dd>	Description

8. HTML Tables

Tables organize information into rows and columns.

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Course</th>
  </tr>
  <tr>
    <td>Rahul</td>
    <td>MCA</td>
  </tr>
</table>
```

Tag	Purpose
<code><table></code>	Creates a table
<code><tr></code>	Table row
<code><th></code>	Table heading cell
<code><td></code>	Table data cell
<code><caption></code>	Table title/caption

9. HTML Forms

Forms are used to collect user input.

```
<form>
  <label>Name:</label>
  <input type="text" name="name">

  <label>Email:</label>
  <input type="email" name="email">

  <input type="submit" value="Submit">
</form>
```

Input Type	Purpose	Example
text	Single-line text	<input type="text">
password	Password input	<input type="password">
email	Email input	<input type="email">
number	Numeric input	<input type="number">
radio	Choose one option	<input type="radio">
checkbox	Choose multiple options	<input type="checkbox">
date	Select date	<input type="date">
file	Choose a file	<input type="file">
submit	Submit form	<input type="submit">
reset	Reset form	<input type="reset">

Other important form elements:

```
<label for="city">City</label>
<select id="city" name="city">
  <option>Delhi</option>
  <option>Mumbai</option>
</select>

<textarea rows="4" cols="30"></textarea>
<button type="submit">Submit</button>
```

★ **Important:** The **name** attribute is commonly used to identify form controls when form data is submitted.

10. HTML Block & Inline Elements

Block elements normally start on a new line and occupy the available width. Examples include **<div>**, **<p>** and headings.

Inline elements normally occupy only the space they need. Examples include ****, **<a>** and ****.

Type	Examples	Basic behavior
Block	div, p, h1, form, section	Starts on a new line
Inline	span, a, strong, em	Stays in the same line

11. Semantic HTML

Semantic elements describe the meaning and role of content, making page structure clearer.

```
<header>Website Header</header>

<nav>Navigation Links</nav>

<main>
  <section>
    <h2>About</h2>
    <p>About content...</p>
  </section>

  <article>Article content...</article>

  <aside>Related content</aside>
</main>

<footer>Copyright</footer>
```

Element	Purpose
<header>	Introductory/header content
<nav>	Navigation links
<main>	Main content of the page
<section>	Thematic section of content
<article>	Self-contained content
<aside>	Related/secondary content
<footer>	Footer information

12. HTML Audio & Video

HTML provides elements for embedding media.

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
</audio>

<video controls width="400">
  <source src="video.mp4" type="video/mp4">
</video>
```

The **controls** attribute displays browser-provided playback controls.

13. HTML Comments

```
<!-- This is an HTML comment -->
<p>Hello</p>
```

Comments are ignored by the browser and are useful for notes inside source code.

14. HTML Entities

Entities allow special characters to be represented safely in HTML.

Entity	Displays
&	&
<	<
>	>
"	"
 	Non-breaking space

15. HTML5 Important Features

Feature	Examples
Semantic elements	header, nav, main, section, article, footer
Multimedia	audio, video
Graphics	canvas, svg
Forms	email, date, number, range, etc.
Storage APIs	Web Storage concepts

16. iframe

The **<iframe>** element can embed another webpage or external content inside a page.

```
<iframe
  src="page.html"
  width="500"
  height="300">
</iframe>
```

17. div and span

```
<div>
  <h2>Product</h2>
  <p>This is a product.</p>
</div>

<p>Hello <span>World</span></p>
```

<div> is a generic container commonly used for grouping block-level content. **** is a generic inline container.

18. HTML Global Attributes

Attribute	Purpose
id	Unique identifier
class	Groups elements
style	Inline CSS styling
title	Additional information
hidden	Hides an element

19. Quick HTML Page Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Page</title>
</head>
<body>
  <header>
    <h1>Student Portal</h1>
  </header>

  <nav>
    <a href="home.html">Home</a>
    <a href="contact.html">Contact</a>
  </nav>

  <main>
    <h2>Welcome</h2>
    <p>This is my HTML page.</p>

    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>
  </main>

  <footer>
    Copyright 2026
  </footer>
</body>
</html>
```

20. Important HTML Interview / MCQ Points

Question / Topic	Answer
HTML	HyperText Markup Language
HTML is a	Markup language, not a programming language
Root element	<html>
Page title	<title> inside <head>
Visible content	<body>
Largest standard heading	<h1>
Paragraph	<p>
Hyperlink	<a>
Image	
Image destination	src attribute
Alternative image text	alt attribute
Unordered list	
Ordered list	
List item	
Table row	<tr>
Table heading	<th>
Table data	<td>
Form	<form>
Single-line text input	<input type="text">
Semantic navigation	<nav>
Main content	<main>
HTML comment	<!-- comment -->

★ Remember:

HTML gives the webpage its **structure**. CSS gives it **style**. JavaScript adds **behavior and interactivity**.

```
HTML      → Structure
CSS       → Style / Design
JavaScript → Behavior / Interaction
```

That's it for HTML Basics! Practice each tag by creating small webpages. **Code + Practice = Skill + Confidence**.